looked at closely sways and trembles, however slightly, so the fields that form the world are subject to minute fluctuations, and it is possible to imagine its basic particles having brief and ephemeral existences, continually created and destroyed by these movements.

This is the world described by quantum mechanics and particle theory. We have arrived very far from the mechanical world of Newton and Laplace, where minute cold stones eternally travelled on long precise trajectories in geometrically immutable space. Quantum mechanics and experiments with particles have taught us that the world is a continuous, restless swarming of things; a continuous coming to light and disappearance of ephemeral entities. A set of vibrations, as in the switched-on hippy world of the 1960s. A world of happenings, not of things.

The details of particle theory were built gradually in the 1950s, 1960s and 1970s, by some of the century's greatest physicists such as Richard Feynman and Gell-Mann. This work of construction led to an intricate theory, based on quantum mechanics and bearing the not very romantic title of 'the Standard Model of elementary particles'. The 'Standard Model' was finalized in the 1970s, after a long series of experiments which confirmed all predictions. Its final confirmation occurred in 2013 with the discovery of the Higgs boson.

But despite the long series of successful experiments, the Standard Model has never been taken entirely seriously by physicists. It's a theory which looks, at least at first sight, piecemeal and patched together. It's made up of various pieces and equations assembled without any clear order. A certain number of fields (but why *these*, exactly?) interacting between themselves with certain forces (but why *these* forces?) each determined by certain constants (but why precisely *these* values?) showing certain symmetries (but again, why *these*?). We're far from the simplicity of the equations of general relativity, and of quantum mechanics.

The very way in which the equations of the Standard Model make predictions about the world is also absurdly convoluted. Used directly, these equations lead to nonsensical predictions where each calculated quantity turns out to be infinitely large. To obtain meaningful results it is necessary to imagine that the parameters entering into them are themselves infinitely large, in order to counterbalance the absurd results and make them reasonable. This convoluted and baroque procedure is given the technical term 'renormalization'. It works in practice but leaves a bitter taste in the mouth of anyone desiring simplicity of nature. In the last years of his life the greatest